CEE 260/MIE 273: Probability and Statistics in Civil Engineering

Lecture 3E: The Poisson Distribution

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Outline

1. Introduction
2. The Poisson distribution
3. Examples
4. Outlook

Recap of Lecture 3b: Binomial distribution

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The above are all **Poisson processes** and the respective probabilities can be modeled by the **Poisson distribution**

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- Number of accidents in an industrial facility over time
- Customer arrivals at a bike store in a given morning

Poisson distribution: brief history

- Attributed to Simeon Denis Poisson (1781–1840), who published it in 1837 in a theory regarding number of wrongful convictions
- Earlier similar work had been published by Abraham de Moivre (in 1711)
- In 1860: Simon Newcomb fitted the Poisson distribution to astronomical data
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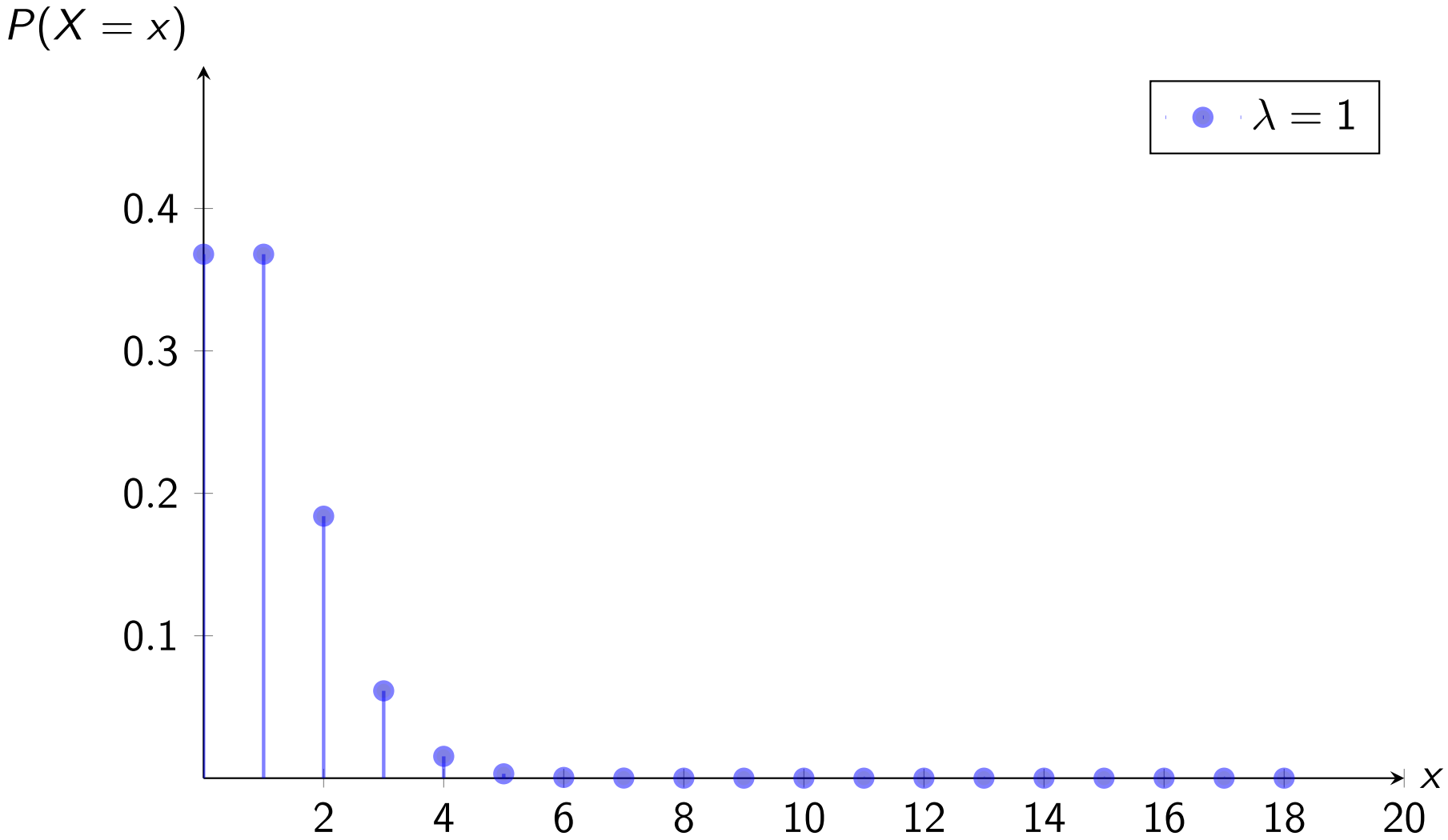
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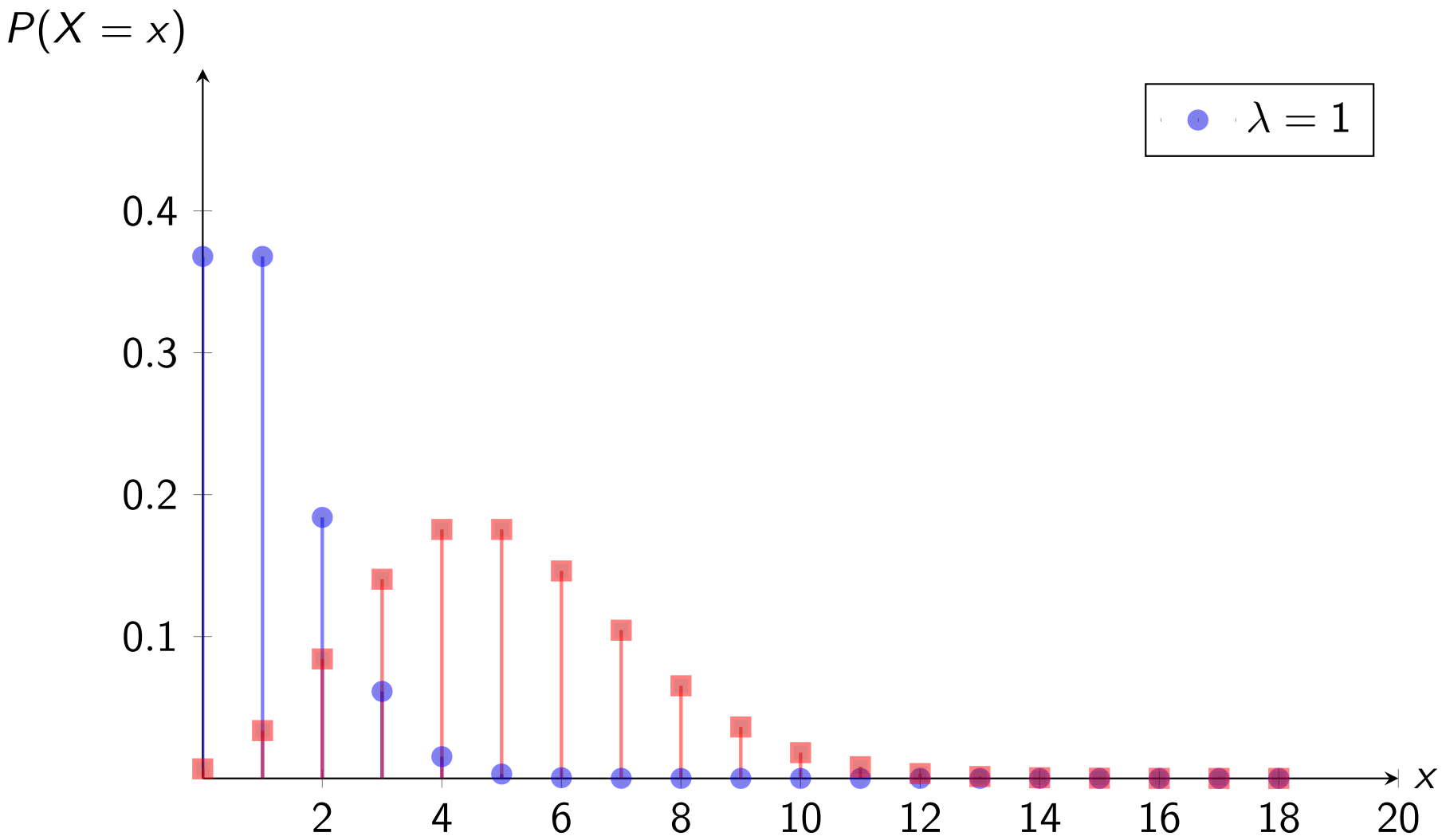
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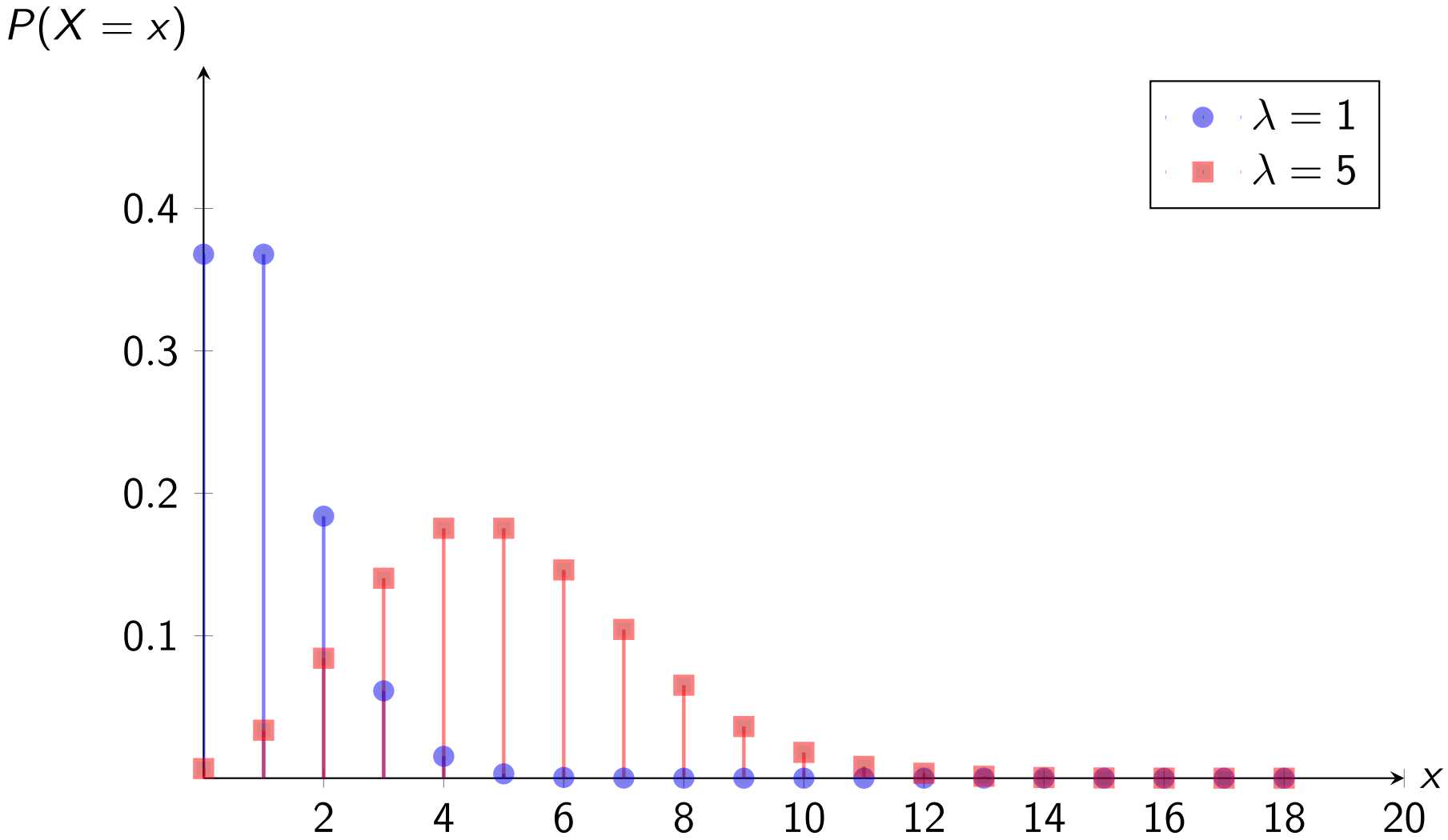
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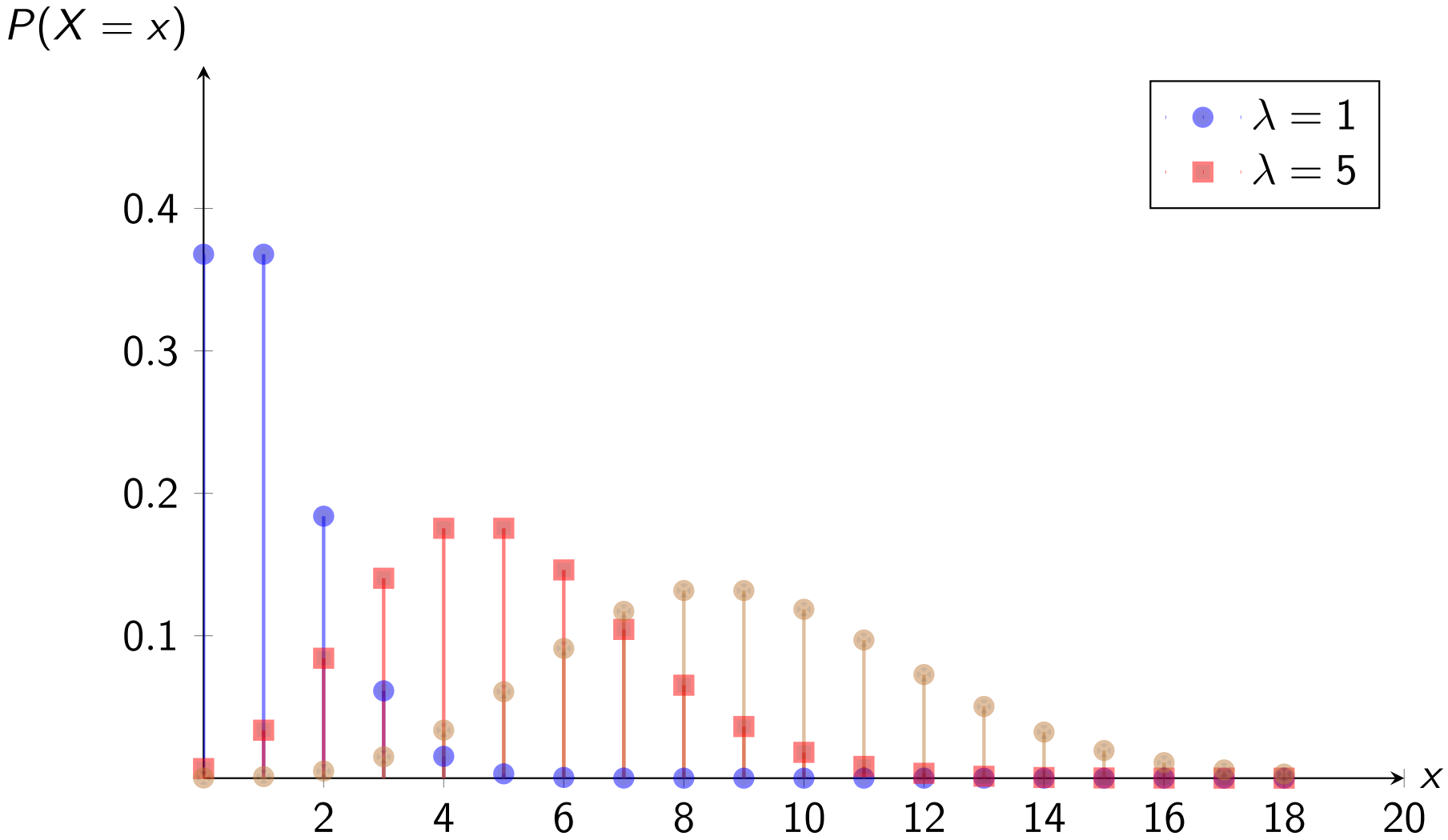
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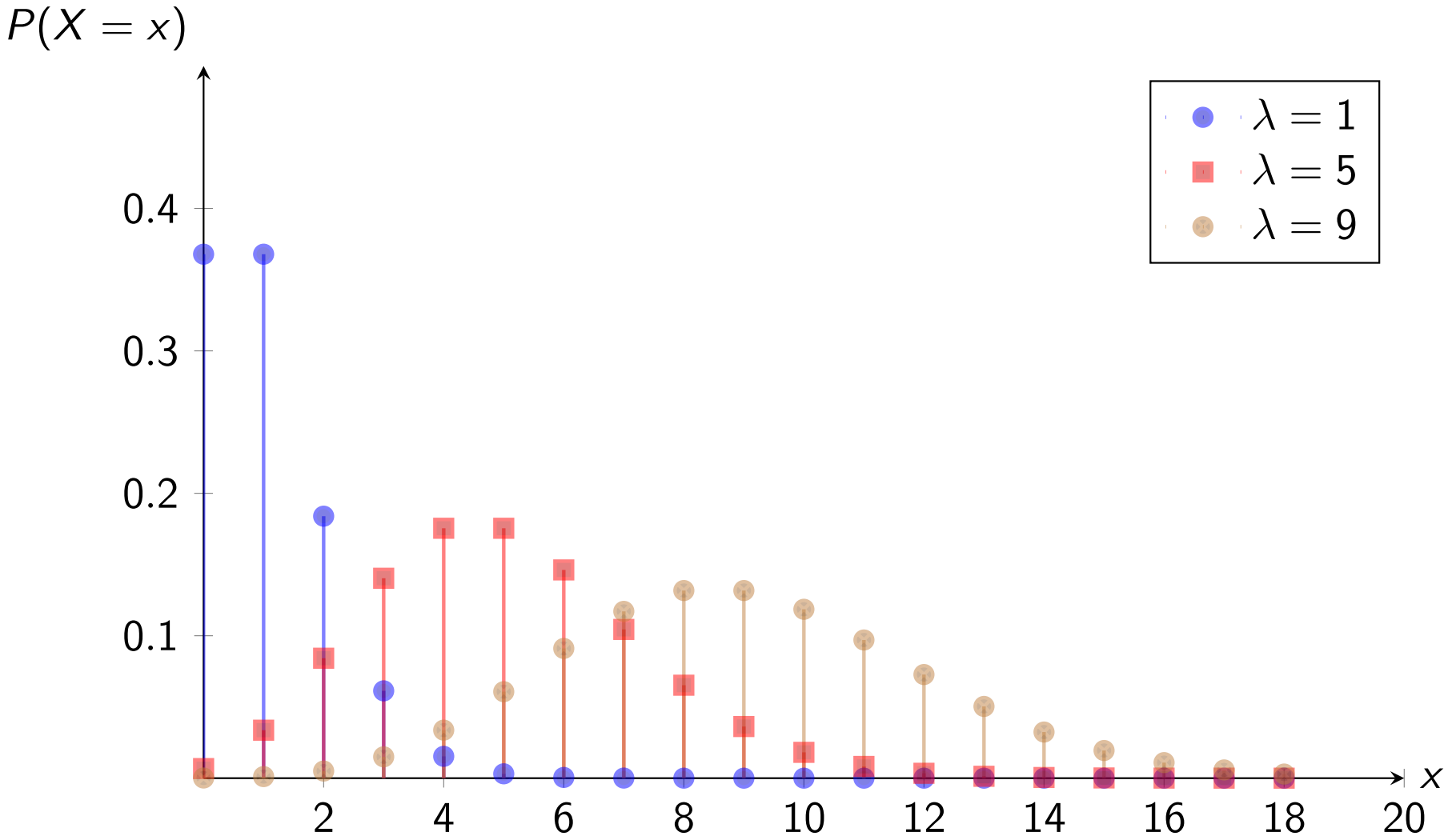
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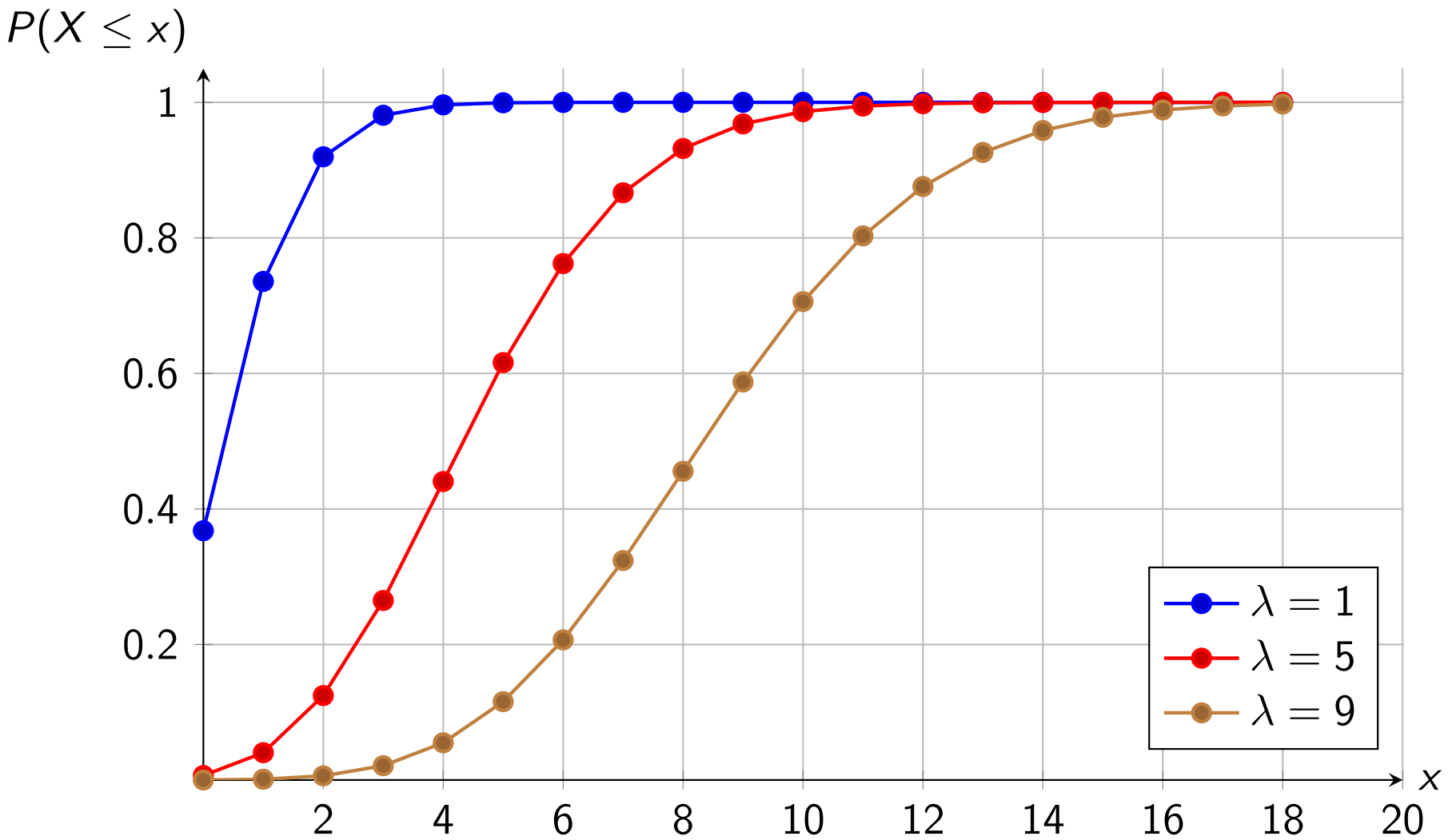
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(a) What is the probability that exactly 6 small aircraft arrive during a 1-hr period?

(b) What is the probability that at least 10 small aircraft arrive during a 1-hr period?

(c) How many aircraft do you expect to arrive during a 90-minute period?

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(a) What is the probability that exactly 6 small aircraft arrive during a 1-hr period?

Solution

$$P(X_{1hr} = 6) = \frac{\lambda^x e^{-\lambda}}{x!} =$$

Example 2: Aircraft arrivals at airport (cont.)

(a) What is the probability that exactly 6 small aircraft arrive during a 1-hr period?

Solution

$$P(X_{1hr} = 6) = \frac{\lambda^x e^{-\lambda}}{x!} = \frac{8^6 e^{-8}}{6!} =$$

Example 2: Aircraft arrivals at airport (cont.)

(a) What is the probability that exactly 6 small aircraft arrive during a 1-hr period?

Solution

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(b) What is the probability that at least 10 small aircraft arrive during a 1-hr period?

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Solution

The desired probability is given by:

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The desired probability is given by:

$$P(X_{1hr} > 9) = 1 - P(X_{1hr} \leq 9)$$

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Manipulating the rate parameter

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- And thus:

$$P(X_t = x) = \frac{(\lambda t)^x}{x!} e^{-(\lambda t)} = \frac{(\lambda')^x}{x!} e^{-(\lambda')} \tag{8}$$

Example 2: Aircraft arrivals at airport (cont.)

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(c) How many aircraft do you expect to arrive during a 90-minute period?

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(c) How many aircraft do you expect to arrive during a 90-minute period?

Solution

Expected arrivals within 90 minutes $= 8/\text{hr} \times 1.5 \text{ hrs} = 12$.

Example 3: Amherst Coffee customer arrivals (revisited)

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Starting at 7 AM, customers arrive Amherst Coffee according to a Poisson process at the rate of 30 customers per hour. Now find the probability more than 65 customers arrive between 10 and 12 PM.

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The probability is now desired for an interval twice as long. So we compute a new rate parameter for 2-hr interval:

$$\lambda^* = 30 \times 2 = 60 \text{ (per 2 hrs)}$$

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Poisson distribution as limit of binomial distribution

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$$X \sim \text{Bin}(n, p) \to X \sim \text{Poisson}(\lambda) \tag{9}$$

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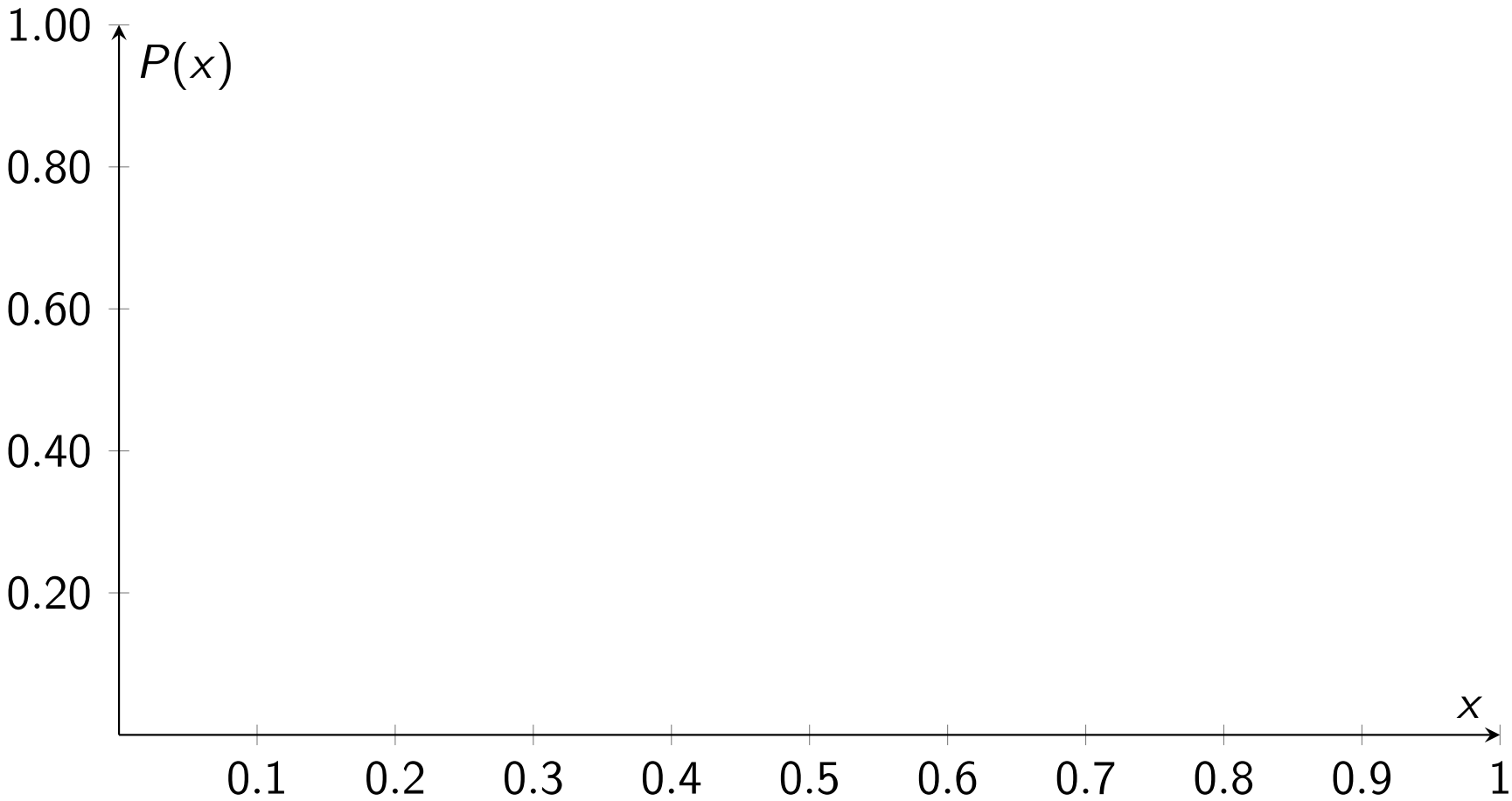
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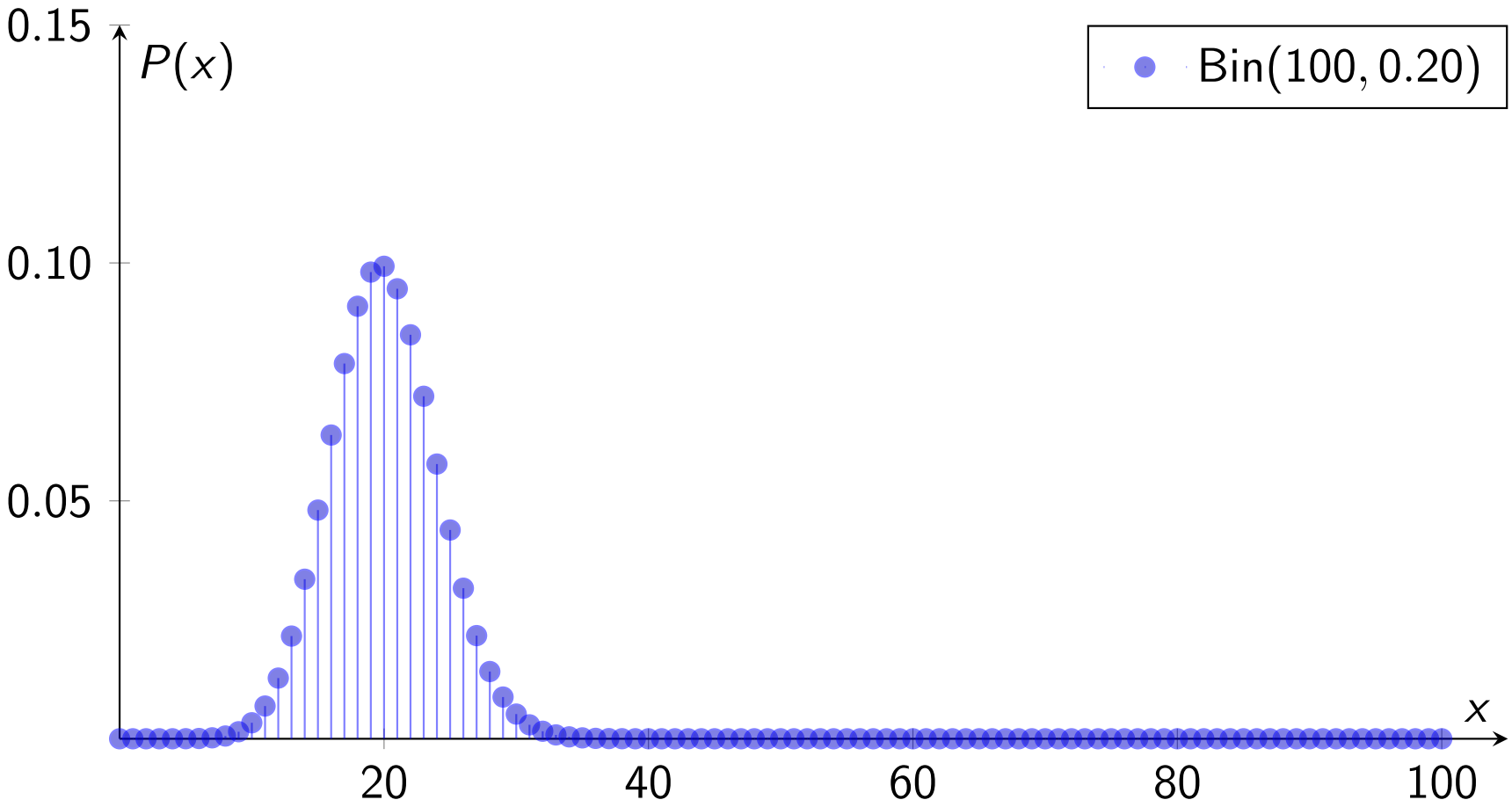


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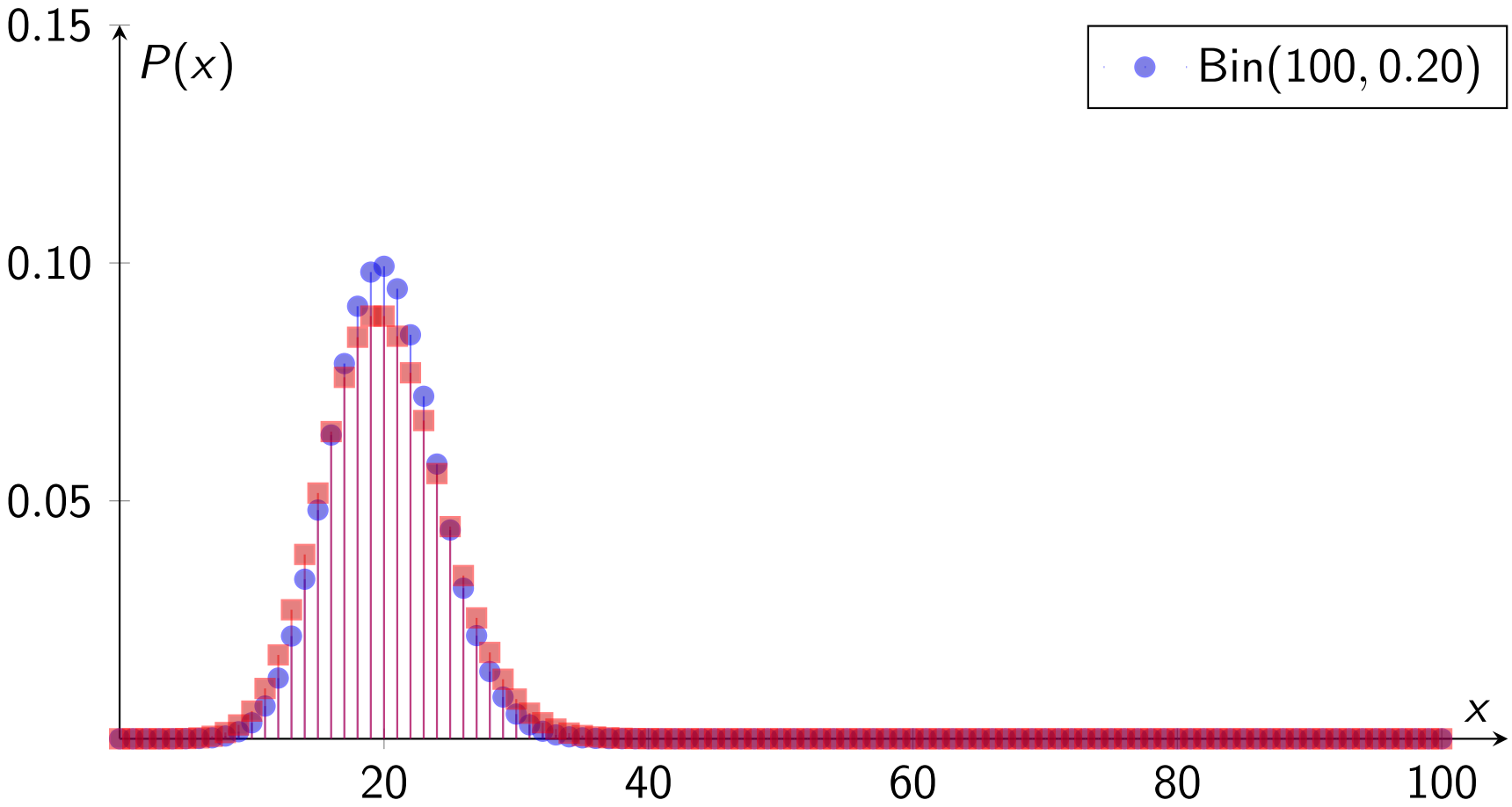


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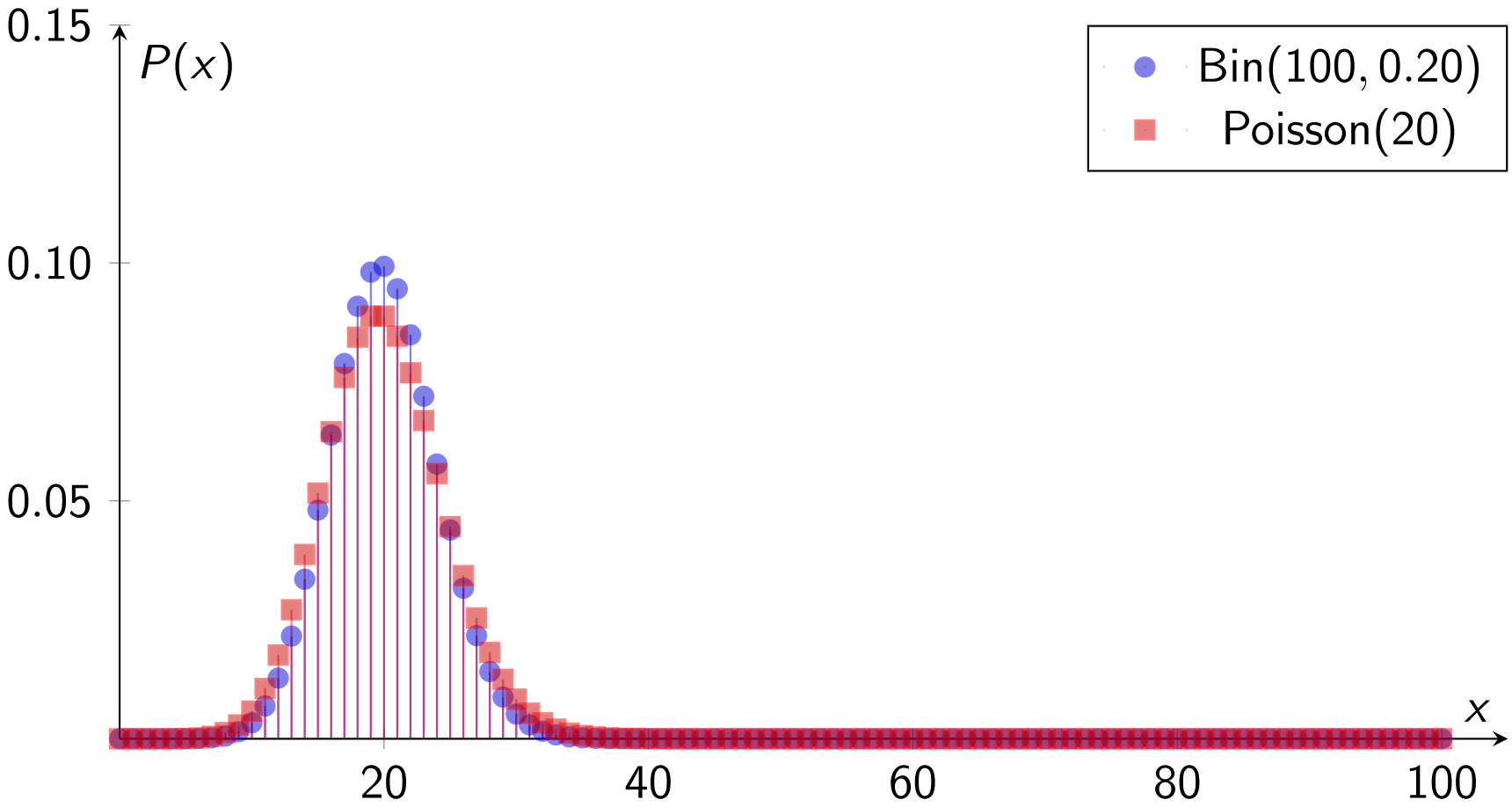


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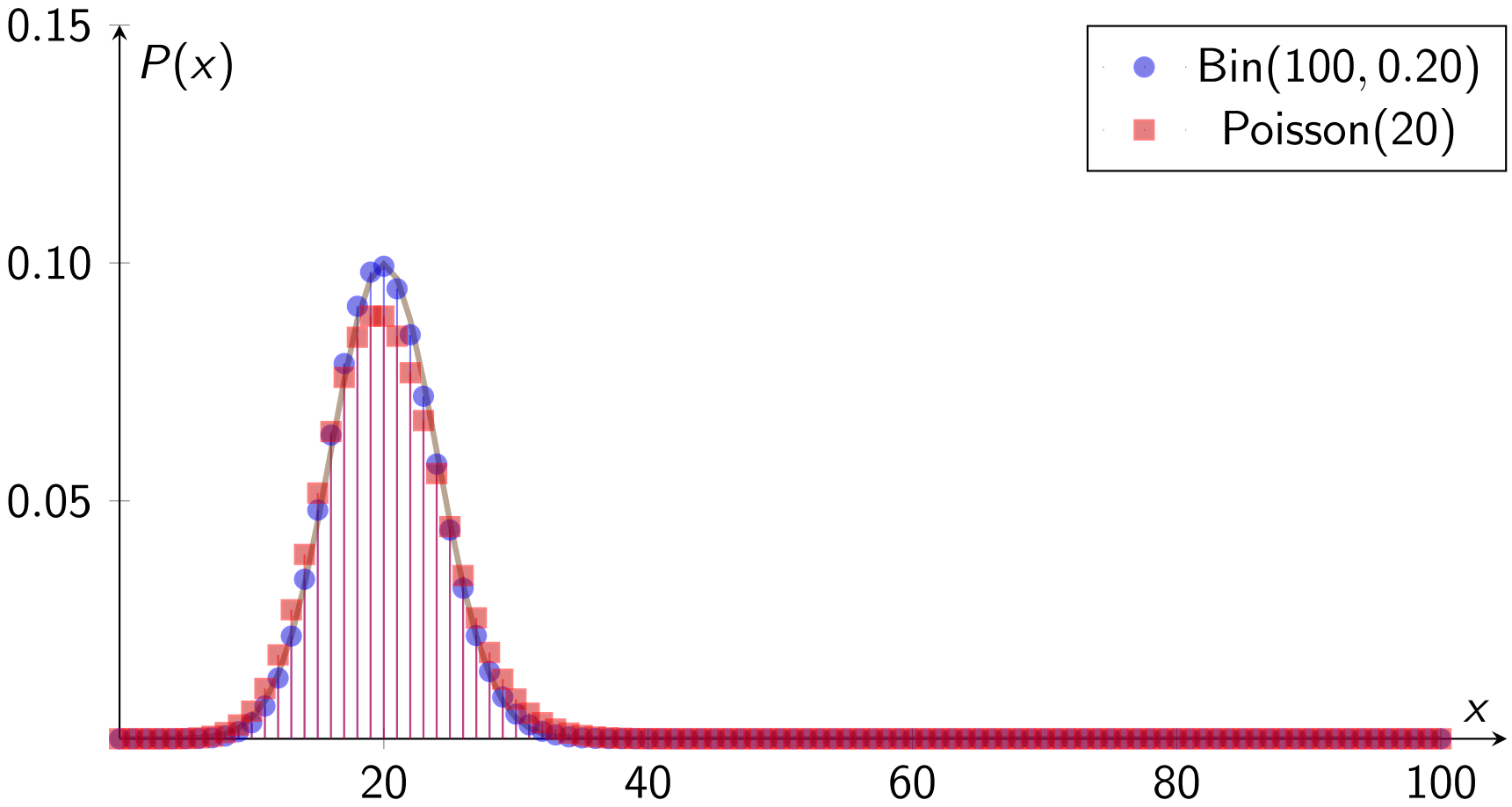


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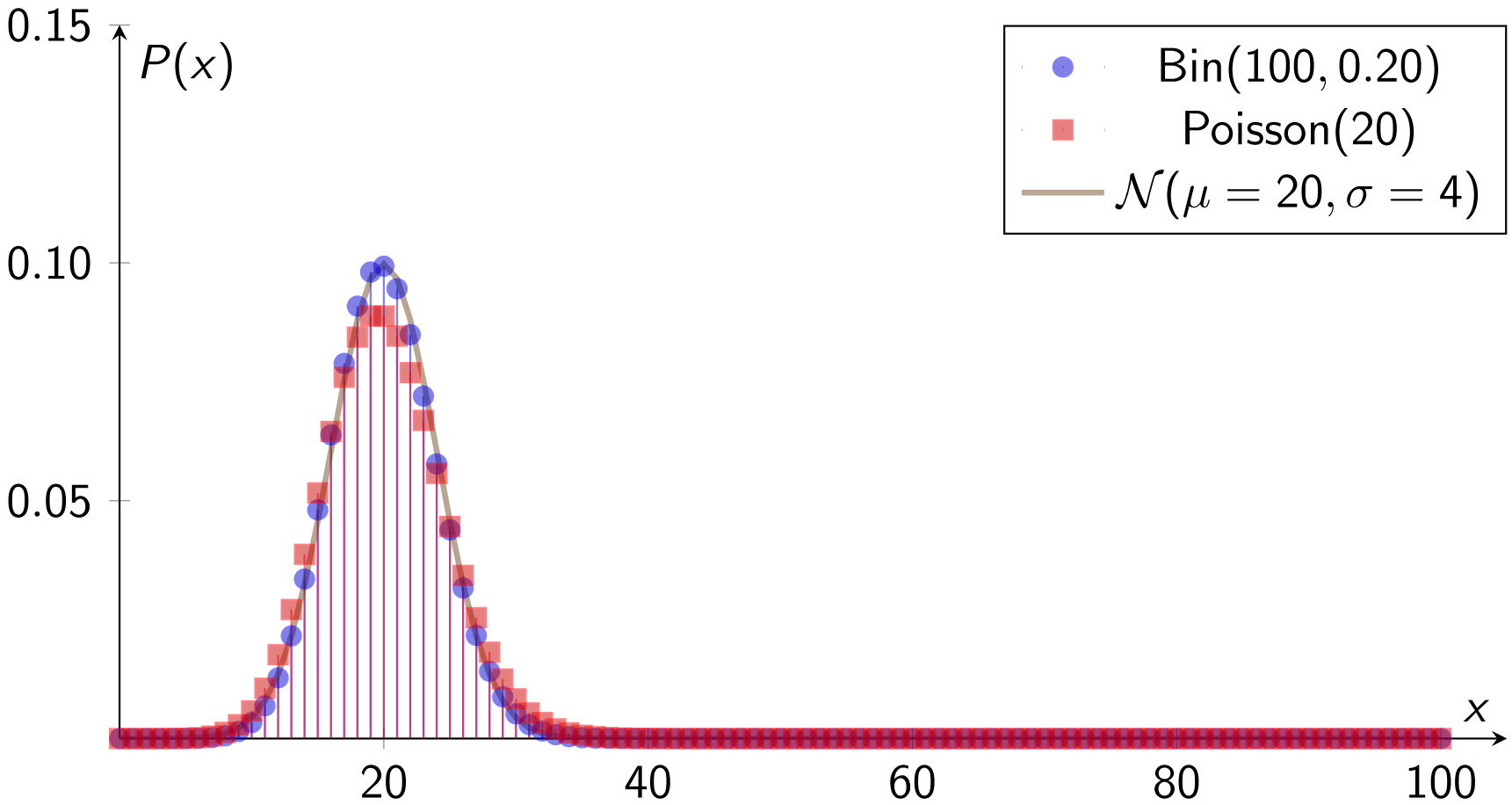


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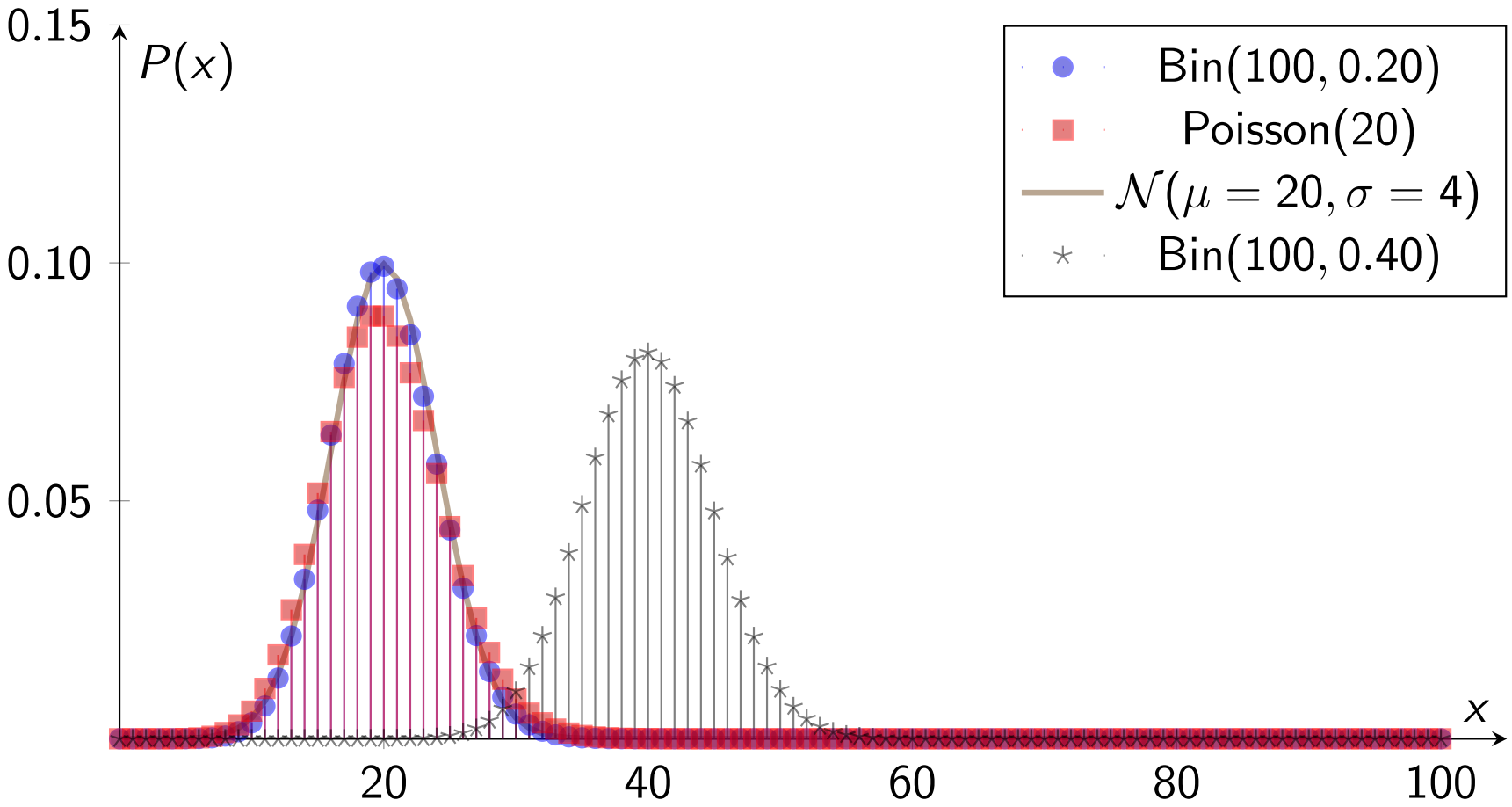


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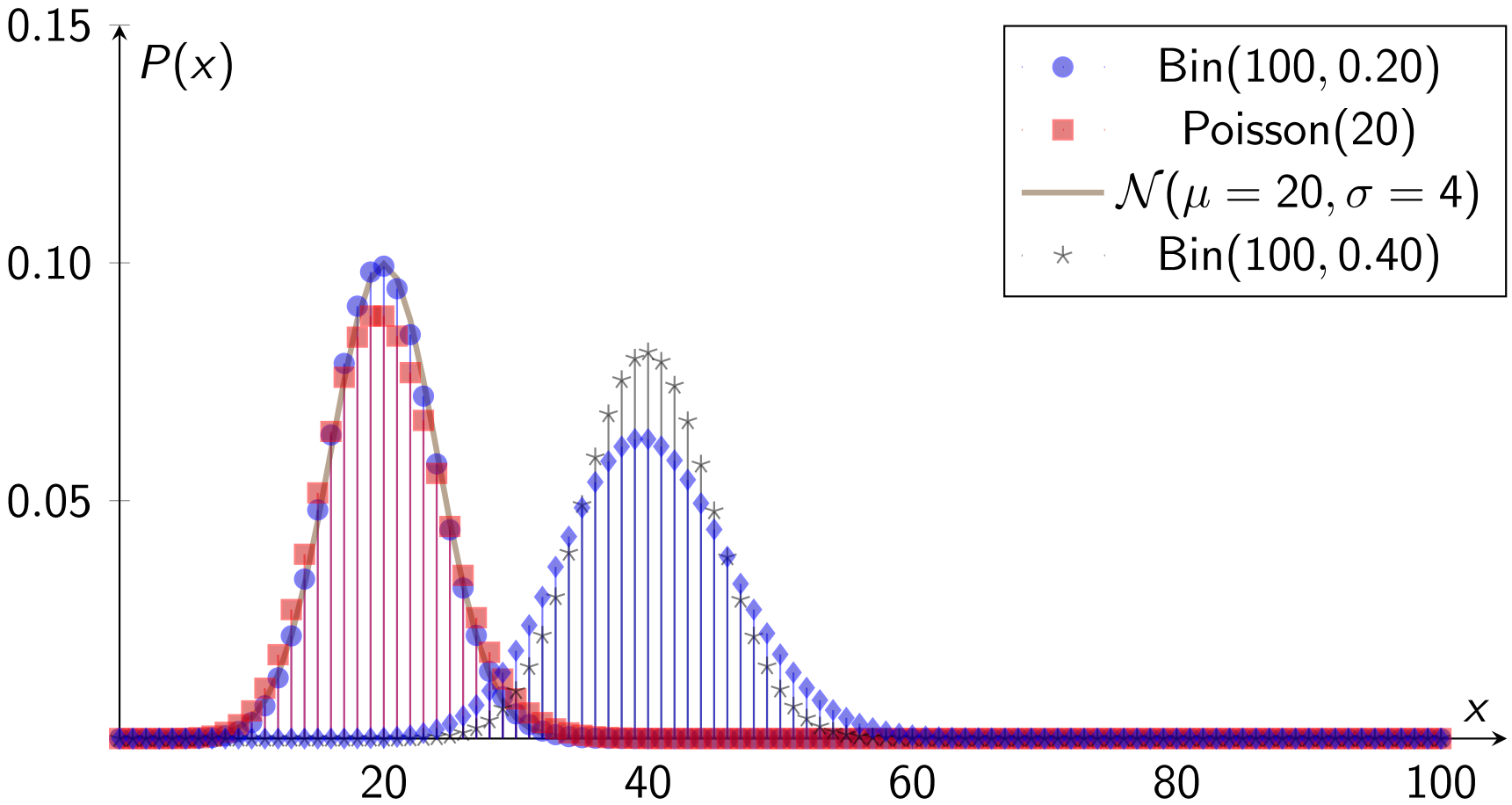


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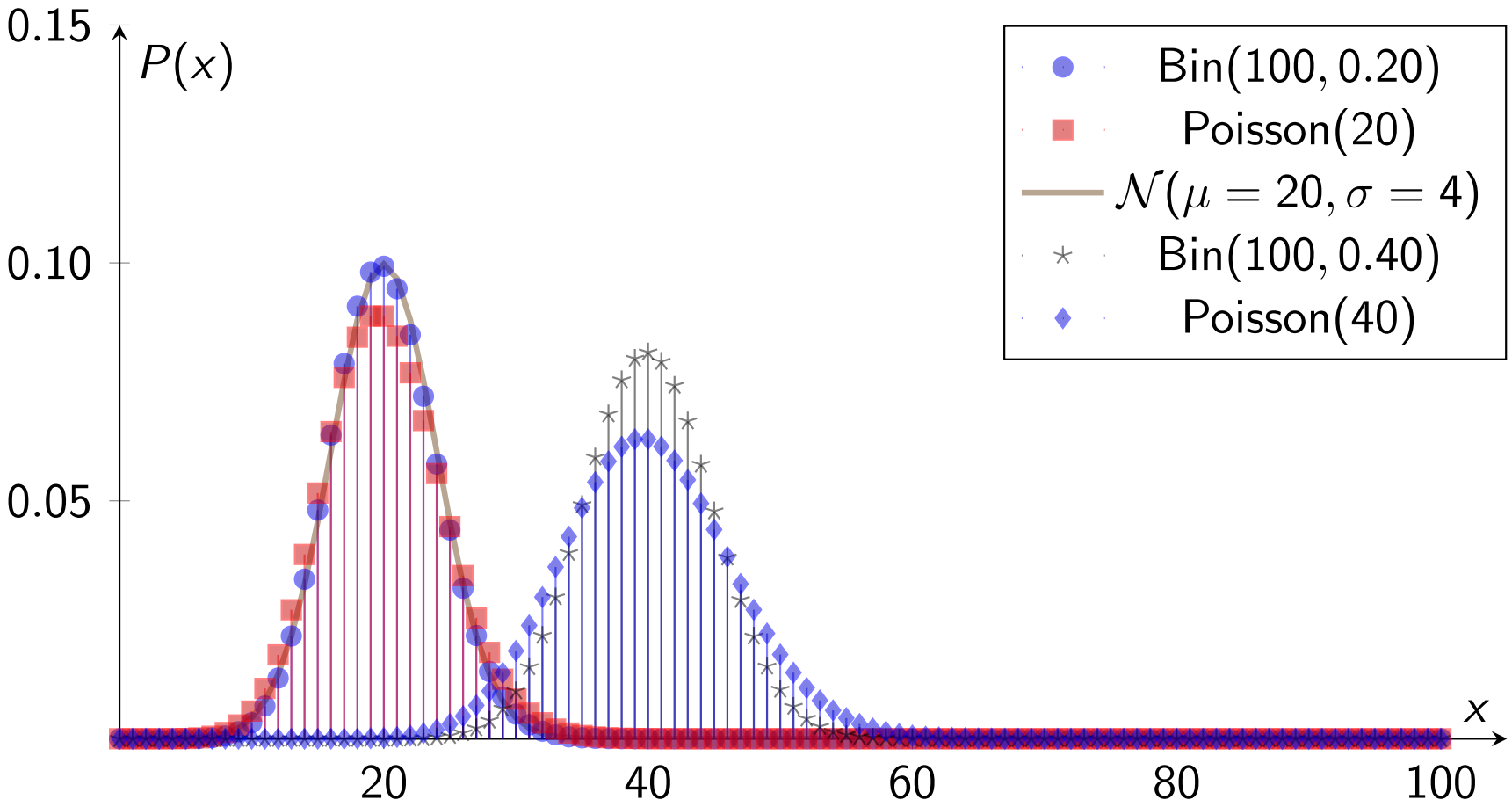


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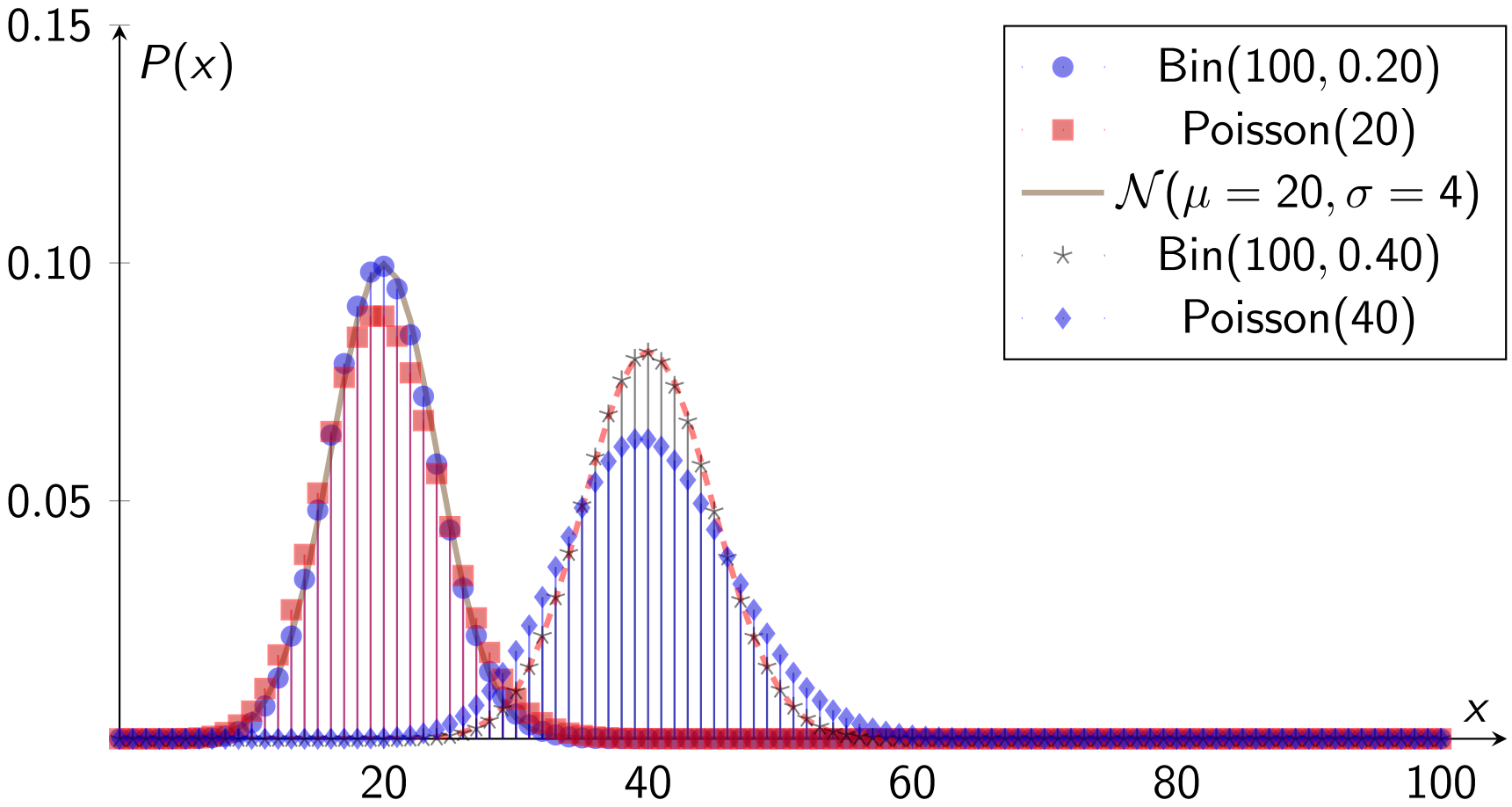


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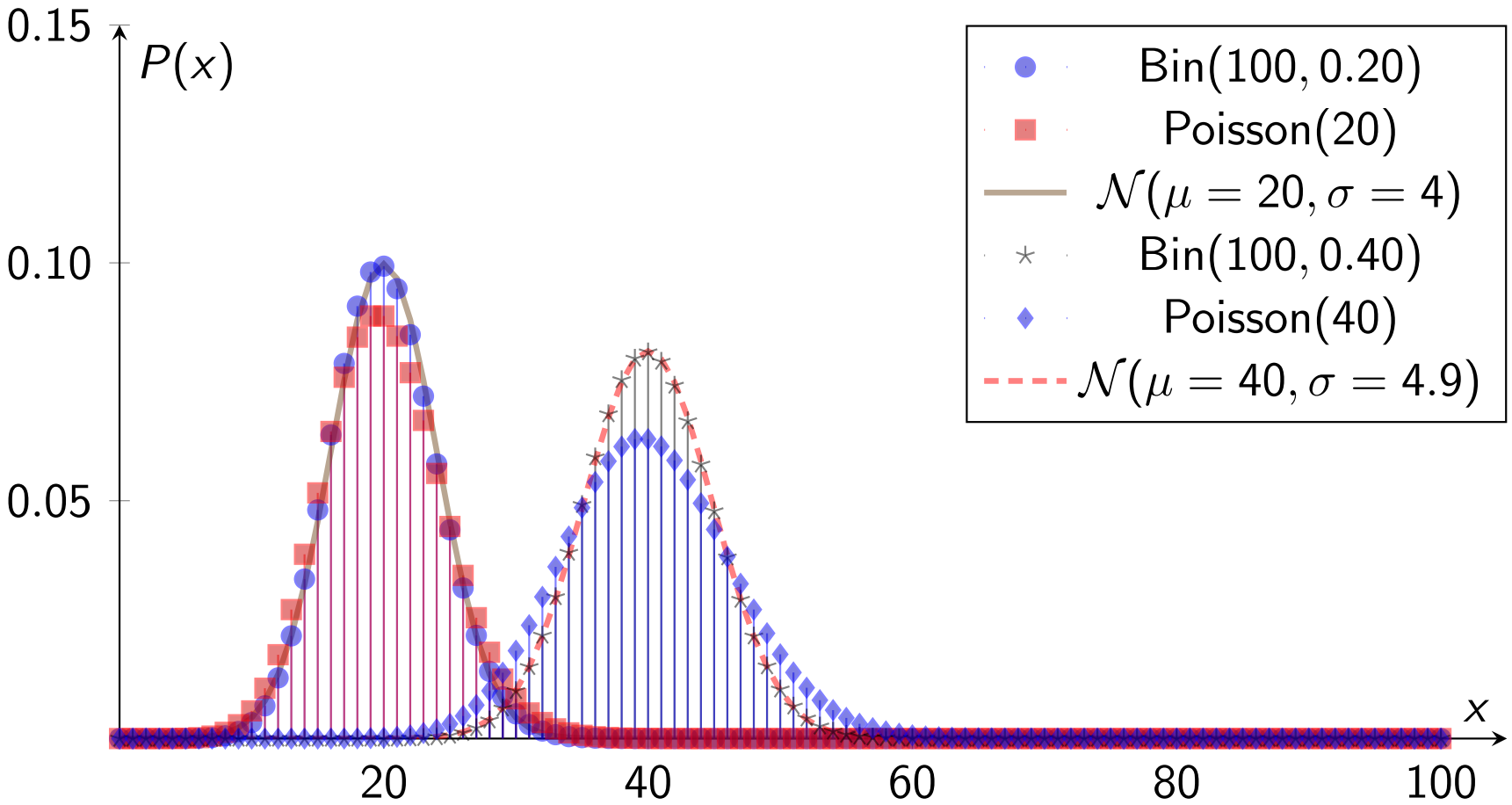


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Reading

- Open Intro Statistics Section 4.5